U.S. County Health Inequality Dashboard

Preliminary Project Report

CSE 564 · Visual Analytics · April 2026 · Naseer Ahmed & Mohtadi Mahim

1. Project overview

This report documents the first working implementation of an interactive visual analytics dashboard for studying health inequality across U.S. counties. The dashboard ties directly to two of the priority domains listed in the project brief — health care and wealth — by surfacing how clinical, economic, and demographic indicators co-vary across roughly 3,000 counties.

The dashboard is a single-page Flask + D3.js application. A Flask backend (Python / pandas / NumPy) loads and serves the dataset and computes a Pearson correlation matrix on demand; the front end is built with D3 v7 and TopoJSON, organized as four linked panels in a 2 × 2 grid. All interactions are client-side after the initial data fetch, so brushing and filtering feel instant.

2. Data and problem

The data is the merged County Health Rankings & Roadmaps file (merged_full_all_states_analysis_dataset.csv) covering all 50 states plus DC. Each row is a county, keyed by 5-digit FIPS, with 20 numeric indicators across four broad themes:

- Health outcomes — life expectancy, years of potential life lost, % fair/poor health, physically/mentally unhealthy days.
- Mortality — drug overdose, firearm fatalities, homicide, and overall injury death rates.
- Access & socioeconomics — primary care physicians per 100k, % uninsured, % unemployed, income ratio, % children in poverty, median household income.
- Behaviors & context — high-school graduation rate, % adults with obesity, % adults smoking, % rural, total population.

The driving question is: which combinations of socioeconomic and behavioral conditions track most strongly with poor health outcomes, and where do those combinations cluster geographically? A linked, multi-view dashboard is the natural tool because it lets the user move between geographic, statistical, and multivariate perspectives on the same set of counties.

3. Implementation in progress

Figure 1 shows the current dashboard composition as it appears on first load. The four panels share a single selection model: any subset of counties highlighted in one view is mirrored in the other three.

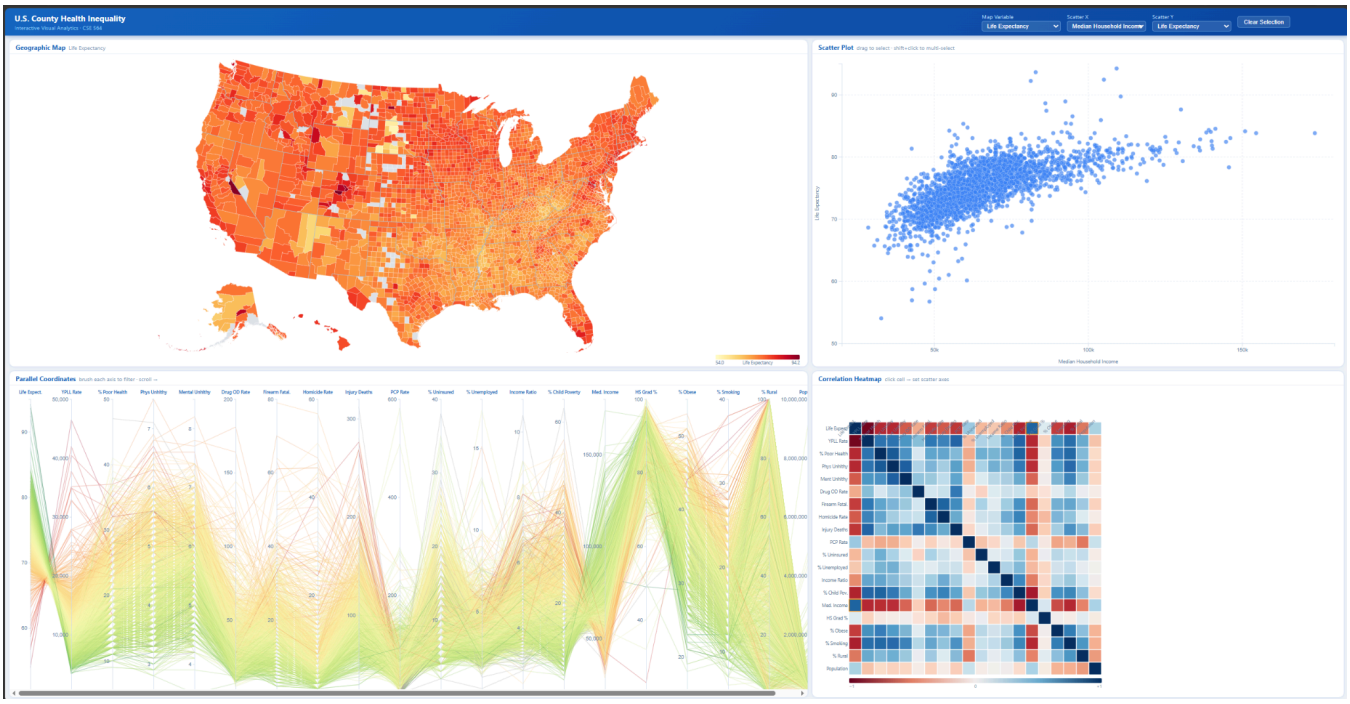

Figure 1 — Current four-panel dashboard: Geographic Map (top-left), Scatter Plot (top-right), Parallel Coordinates (bottom-left), Correlation Heatmap (bottom-right).

4. Panel walkthrough

4.1 Geographic Map (choropleth)

A county-level choropleth renders all FIPS-coded counties using a TopoJSON U.S. atlas. The color encoding is driven by the dropdown “Map Variable,” defaulting to Life Expectancy. Hovering a county reveals a tooltip with state, county, and value; clicking a county adds it to the global selection. Alaska and Hawaii are positioned as small inset frames on the lower-left.

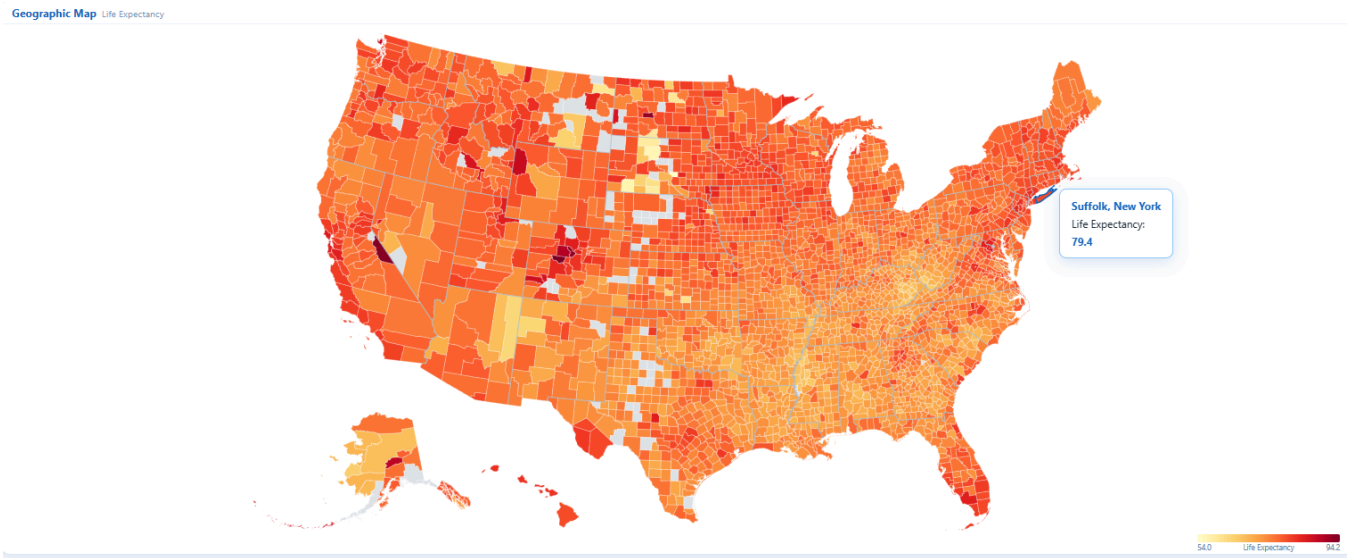

Figure 2 — Geographic map panel (state-level summary view shown for the report). The deployed app draws all ~3,100 counties with a continuous yellow→red colormap.

4.2 Scatter Plot

The scatter plot shows two user-chosen variables (defaults: Median Household Income on X, Life Expectancy on Y). It supports drag-rectangle brushing and shift-click multi-select. The axis pair can also be set indirectly by clicking a cell in the correlation heatmap, which is one of the linkages we are most pleased with.

Figure 3 — Scatter plot showing the strong positive relationship between household income and life expectancy across U.S. counties.

4.3 Parallel Coordinates

The parallel-coordinates view stacks all 20 numeric indicators on independent vertical axes. Each polyline is a single county; line color encodes life expectancy on a red-to-green scale. Each axis supports D3 brushing — dragging a brush filters the dataset live, and the filtered set propagates back to the map, scatter, and heatmap.

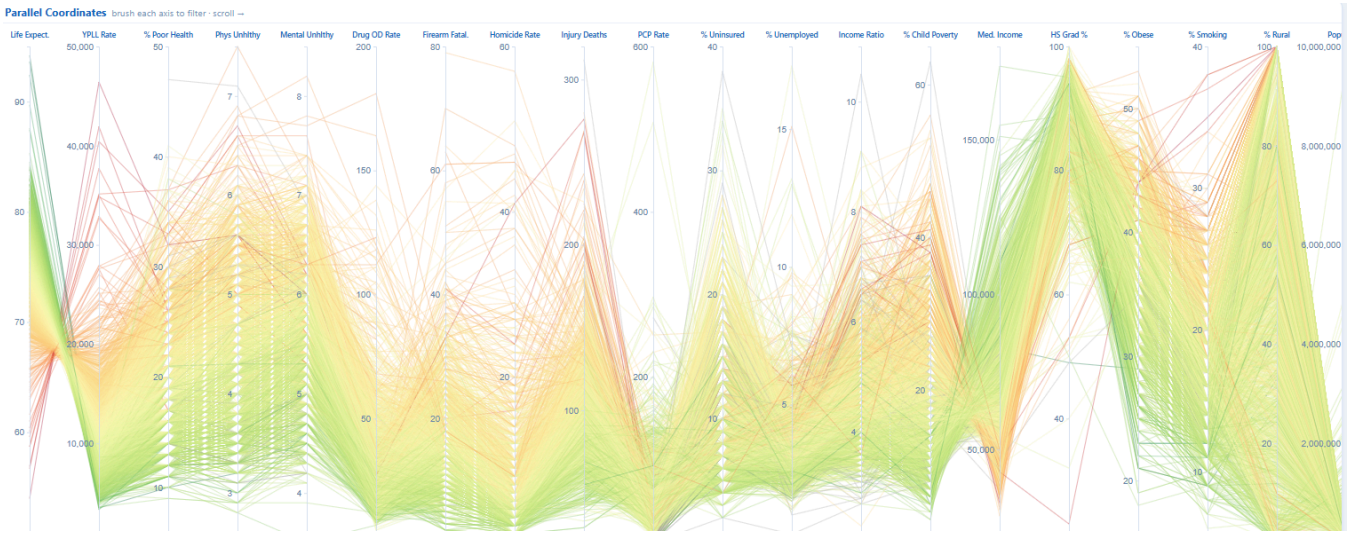

Figure 4 — Parallel coordinates over the first ten indicators, color-coded by life expectancy. The full deployed view scrolls horizontally across all 20 axes.

4.4 Correlation Heatmap

The heatmap presents the Pearson correlation matrix of all 20 numeric indicators (the matrix is computed once on the server in /api/correlations and cached on the client). Cells use a diverging red→white→blue scale. Clicking a cell drives the scatter plot's axes to that variable pair, turning the heatmap into a fast-pivot exploration tool.

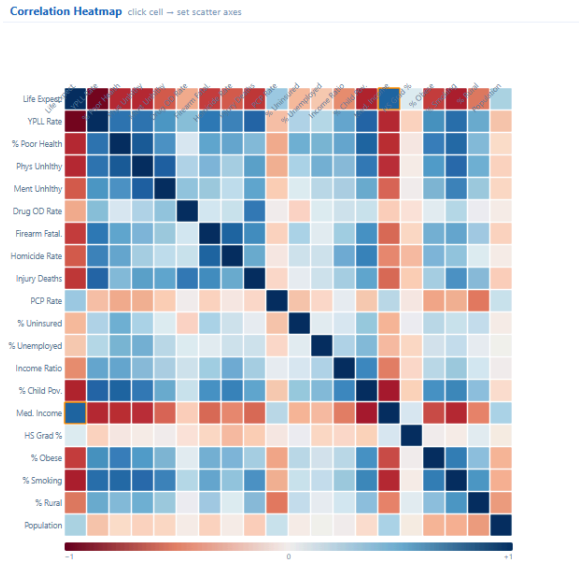

Figure 5 — Correlation heatmap of the 20 indicators. Strong positive blocks (e.g., poor-health metrics) and strong negative blocks (e.g., income vs. unhealthy days) are immediately visible.

5. Linked interactions

All four panels share a single client-side selection set keyed by county FIPS. The interaction graph implemented so far:

- Map → Scatter, Parallel, Heatmap — clicking a county or lasso-selecting a region filters every other panel.

- Scatter → Map, Parallel — drag-rectangle brushing highlights counties in the map and dims non-selected polylines.
- Parallel → Map, Scatter — axis brushes filter the global selection.
- Heatmap → Scatter — clicking a cell sets the scatter X/Y dropdowns to that variable pair and re-renders.
- Header dropdowns — Map Variable, Scatter X, Scatter Y can be changed at any time without losing the current selection.
- Clear Selection button — resets the global selection set and restores all views.

6. Requirements compliance

The implementation already satisfies the requirements from the project brief:

Requirement	Status	How it is met
Real-world problem	**Met**	U.S. county-level health inequality — disparities in life expectancy, mortality, and access to care.
Real dataset	**Met**	County Health Rankings & Roadmaps merged dataset — 3,000+ counties, 20 numeric indicators.
Persuasive boost	**Met**	Tied to health care, wealth, and education — three of the listed high-stakes domains.
Mix of standard + non-standard charts	**Met**	Standard: scatter plot. Non-standard: choropleth geo-map, parallel coordinates, correlation heatmap.
Single-screen, no scrolling	**Met**	2 × 2 grid laid out via CSS grid; all four panels fit on one screen at common resolutions.
Linked brushing across views	**Met**	Selections on the map, scatter, or parallel coordinates highlight the same counties everywhere; heatmap cells drive scatter axes.
Interactive analytics	**Met**	Pearson correlation matrix computed server-side; brush filters and dropdown variable swaps re-render in real time.

7. Next steps before final delivery

- Tighten responsive layout so all four panels remain visible at 1366×768 without truncation.
- Add a small bar-chart panel ranking the top-N counties on the active map variable inside the current selection.
- Cache the topojson + data fetch and add a 1-line legend per panel for accessibility.
- Author a short interactive “story” walkthrough (3 saved selections) for the demo video.
- Record YouTube demo, finalize zip of code + data, prepare slides for the May 12 Zoom presentation.
- Fill in missing data (e.g., imputation or defaults).
- Improve rendering performance (optimize point drawing, reduce re-renders, use memoization/canvas where needed).

Repository: health-inequality-dashboard · Stack: Flask + pandas + D3.js + TopoJSON